

SMART CAMPUS COMPANION

Analysis & Requirements Document

Multi-Platform Software Development for Real-World Problems (CSE)

Problem 1: Smart Campus Companion

Research Method: Online survey (Google Forms) distributed via WhatsApp and Telegram

Total Respondents: 13 (10 Students + 3 Staff Members)

Date: May 20, 2026

Git Repository link:

<https://github.com/Abukiya/smart-campus-companion>

1. Introduction

This document presents the findings from a user research study conducted for the Smart Campus Companion mobile application. The research was carried out during the final examination period using an online survey distributed through Telegram groups to reach both students and staff without disrupting their schedules.

The study investigates how students and staff currently access, share, and receive campus information — and identifies the critical failures in the existing communication system that a mobile application can solve.

1.1 Project Objective

To design and develop a mobile application that centralizes campus information delivery, replacing the fragmented and unreliable informal channels currently used by students and staff, and enabling direct, real-time communication between the university and its members.

1.2 Scope

- Platform: Mobile application (Flutter — primary).
- Target Users: Students and university staff (academic and administrative)
- Core Problem: Campus information fragmentation leading to missed classes, quizzes, and examinations

2. Research Methodology

Given that data collection occurred during the final examination period, in-person interviews were not feasible. The team adapted by deploying an online survey accessible to both students and staff through a single link, using branching logic to present role-appropriate questions.

2.1 Methods Used

Method	Description
Online Survey (Primary)	A Google Form with branching questions distributed via Telegram class groups. Collected 13 responses passively within hours.
Role-based Branching	Q1 (Student vs. Staff) automatically routed respondents to separate question sets, ensuring relevant data from both groups in one form.
Open-ended Questions	Paragraph-type questions captured real stories, frustrations, and specific incidents that quantitative data alone cannot reveal.

2.2 Respondent Profile

13	10	3	100%
Total Responses	Students	Staff Members	Students in 3rd Year

Student departments represented: Computer Science & Engineering (7), Civil Engineering (1), Chemical Engineering (1), Other (1)

Staff roles: 2 Lecturers/Academic Staff, 1 Department Secretary/Administrator

3. Key Findings

3.1 How Students Currently Receive Information

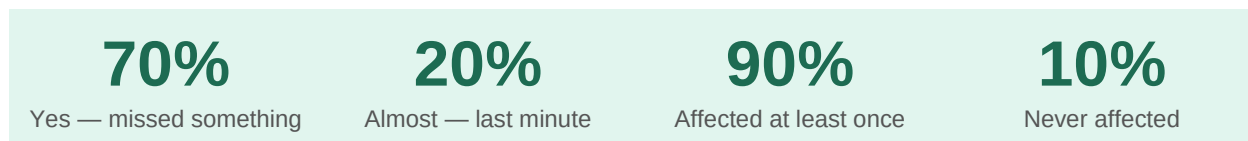
Students were asked to select all channels they use to find out about schedule changes, room changes, or cancellations:

Information Channel	Usage Rate
Telegram group	9 out of 10 students (90%)
A classmate tells me	4 out of 10 students (40%)
Physical notice board	2 out of 10 students (20%)
Show up and find out on the spot	1 out of 10 students (10%)
University website or student portal	0 out of 10 students (0%)

Critical observation: Not a single student uses the official university website or portal as an information source. Unofficial channels Telegram groups managed by student representatives have entirely replaced formal systems.

3.2 Missed Information — Scale of the Problem

Students were asked whether they had ever missed a class, exam, or important event due to not receiving information in time:



9 out of 10 students have been affected by information failures at least once. The consequences reported include failed quizzes, missed examinations, and in one documented case, a grade of F.

3.3 Difficulty Finding Offices and Contact Information

Students rated the ease of finding a specific office, room, or building on campus on a scale of 1 (Very difficult) to 5 (Very easy):

Metric	Value
Average Score	2.3 out of 5
Score of 1 (Very difficult)	2 students
Score of 2	4 students
Score of 3	3 students
Score of 4	1 student
Score of 5 (Very easy)	0 students

A score of 2.3 out of 5 confirms that finding offices and rooms on campus is closer to 'very difficult' than 'easy' for the majority of students. No student rated it a 5.

3.4 Most Wanted App Features

Students selected which features they would actually use if a campus mobile app existed:

Feature	Demand
Personal class timetable	6/10 students (60%)
Real-time alerts for room/cancellation changes	5/10 students (50%)
Staff and office contact directory	5/10 students (50%)
Exam schedule and academic calendar	5/10 students (50%)
Official announcements and news feed	4/10 students (40%)
Campus map with building and room locator	2/10 students (20%)

One student additionally requested offline functionality: 'I need an offline app that will tell me if I have a class.' This directly supports implementing an offline-first architecture as an innovation feature.

4. Identified Pain Points

Based on synthesis of open-ended survey responses, five major pain points were identified:

Pain Point 1 No Centralized Information Source

Students rely on multiple fragmented Telegram channels. There is no single official place to check. If a student is not in the correct group or if the group admin is inactive — they receive no information at all.

"There is no one place that you can go to get the information — there are different telegram channels but they are not centralised." — 3rd Year CSE Student

Pain Point 2 Over-Reliance on Class Representatives

All 10 students rely on representatives or classmates as intermediaries. This creates a critical single point of failure: if the representative misunderstands the message, delays forwarding it, or the student is not reachable, the information never arrives.

"Missed a quiz due to misunderstanding of the representative text about the time of the quiz." — 3rd Year CSE Student

Pain Point 3 Surprise Assessments With No Digital Record

Multiple students reported being quizzed or assessed without adequate prior notice. Lecturers verbally announce assessments in class, creating no persistent digital record. Students absent from that precise moment have no way of knowing.

"The teacher quizzed the students on the spot without telling us we had class that day." — 3rd Year CSE Student

Pain Point 4 Inaccessible Staff Contact Information

Students cannot directly contact lecturers or staff. They must go through class representatives or physically visit department offices. Phone numbers and emails are not publicly accessible.

"Almost — we cannot get the phone numbers." — 3rd Year CSE Student

Pain Point 5 — Severe Real-World Consequences

Information failures are not minor inconveniences. Reported consequences include: missed quizzes, missed examinations, a grade of F, and repeated near-misses. The system fails regularly enough that 9 out of 10 students have been impacted.

"Grade F." — 3rd Year Chemical Engineering Student — describing the result of missing information

5. Staff Research Findings

5.1 Current Communication Methods

All 3 staff members rely on class representatives as their primary channel. None use a direct digital channel that reaches all students reliably.

Communication Channel	Usage
Through class representatives	All 3 staff (100%)
Verbal announcement in class	2 staff (67%)
Telegram group	1 staff (33%)
Physical notice board	1 staff (33%)
Email to students	0 staff (0%)
University website or portal	0 staff (0%)

5.2 Where Staff Say Communication Breaks Down

Staff selected all that apply from a list of communication failure points:

- Information doesn't reach students fast enough — selected by all 3 staff
- No single reliable channel everyone uses — selected by all 3 staff
- Students don't check official channels — selected by 2 staff
- Information gets lost between departments — selected by 1 staff

5.3 Staff Acknowledgment of the Problem

"I am so tight with different works that I can't give information on time to students." — **Lecturer**

"Sometimes it is not easy to communicate through a representative. They might not pick up the phone." **Department Secretary**

"Information not given early enough." **Lecturer**

5.4 Staff Willingness to Adopt a Solution

All 3 staff members responded 'Yes, if it's simple enough' when asked whether they would use a simple admin dashboard to push announcements directly to students' phones. No staff member refused.

"A centralized system where I can get all students from different departments to share the needed information." **Lecturer describing their ideal solution**

6. As-Is Workflow (Current State)

The following describes how campus information currently flows — and fails — based on the research findings:

Step 1: Information is created

A lecturer decides to change a room, cancel a class, or schedule a quiz. This decision exists only in their head or in a private note.

Step 2: Verbal or informal relay

The lecturer informs the class representative verbally — either in person, by phone call, or through a personal Telegram message. There is no official system for this step.

Step 3: Representative broadcasts to a group

The class representative forwards the message to a Telegram group. The message may be paraphrased, misunderstood, or delayed. Students not active in the group at that moment may miss it.

Step 4: Student-to-student relay (informal)

Students who receive the message may pass it on to friends who missed it. This chain is unreliable and incomplete — some students are never reached.

Step 5: Discovery at the point of failure

Students who were not reached find out by arriving at the wrong room, finding an empty classroom, or sitting an unannounced quiz they did not know about.

Failure points: Step 2 (no official channel), Step 3 (representative as single point of failure), Step 4 (uncontrolled relay). Any one of these can sever the information chain completely.

7. Real-Life Student Experience Scenarios

The following scenarios are constructed directly from real incidents described by survey respondents:

Scenario 1 : The Quiz No One Knew About

A 3rd-year CSE student received a message from their class representative about an upcoming quiz. The representative had misread the time from the lecturer's message. The student arrived at the correct time according to the representative's message — which was wrong. The quiz had already ended. No official record of the announcement existed. The student received a zero.

Scenario 2 : Waiting for a Call That Never Came

A student's informal system for confirming whether class was happening was to wait for a friend to call and confirm. On this occasion the friend forgot to call — not because there was no class, but because they assumed the student already knew. The student stayed home, assuming there was no class. There was. They missed it.

Scenario 3 : Wrong Day, Lost Quiz

A student saw a Telegram message from their group about an upcoming quiz. Reading quickly, they assumed the quiz was the following day. Without a personal timetable app or push notification to confirm, they did not double-check. The quiz was that same day. They missed it entirely.

Scenario 4 : Grade F from a Missed Exam

A Chemical Engineering student missed an examination because the information about the exam date or venue change did not reach them. No push notification was sent, no official channel existed, and the student was not in the active WhatsApp group at the time. The result was a grade of F.

Scenario 5 : The Invisible Announcement

A lecturer verbally announced at the end of one class that a quiz would be held in the next session. Students who were present heard it. No digital record was created — no message was sent, no post was made. Students who arrived late to that class, or who were absent, had no way of knowing. Several arrived to the next class unprepared or not at all.

8. Requirements Specification

8.1 Functional Requirements

The following functional requirements are derived directly from the research findings:

ID	Requirement
FR-01	The system shall display a personalized student dashboard showing today's schedule, upcoming classes, and active announcements upon login.
FR-02	The system shall allow students to view their full class timetable and academic calendar, including exam dates and submission deadlines.
FR-03	The system shall send real-time push notifications to students when a class is cancelled, a room is changed, or a new announcement is posted.
FR-04	The system shall provide an official announcements feed where staff can publish updates and students can view them in chronological order.
FR-05	The system shall include an interactive campus map showing buildings, offices, and rooms with search functionality.
FR-06	The system shall provide a staff and office contact directory searchable by name, department, or role.
FR-07	The system shall provide an admin dashboard for staff to post announcements, schedule changes, and cancellations directly to students.
FR-08	The system shall support offline access to the student's last-synced timetable and announcements when no internet connection is available.
FR-09	The system shall support user authentication (student login and staff login) with role-based access control.
FR-10	The system shall maintain a notification history so students can review past alerts they may have missed.

8.2 Non-Functional Requirements

ID & Category	Requirement
NFR-01 — Performance	The app shall load the student dashboard within 2 seconds on a standard mobile connection.
NFR-02 — Reliability	Push notifications shall be delivered within 30 seconds of being posted by a staff member.
NFR-03 — Offline Support	The app shall display the last-synced timetable and announcements without an internet connection (innovation feature).
NFR-04 — Usability	The admin interface for staff must require no more than 3 steps to post an announcement — staff have limited time.
NFR-05 — Security	Student and staff data shall be protected. Authentication tokens shall

	expire after 24 hours of inactivity.
NFR-06 — Compatibility	The app shall run on Android 8.0 and above and iOS 13 and above.
NFR-07 — Scalability	The system shall support at least 500 concurrent users without performance degradation.

9. Use Cases & User Stories

9.1 Student User Stories

ID	User Story
US-01	As a student, I want to see today's class schedule on my dashboard so I know exactly where to be and when.
US-02	As a student, I want to receive a push notification immediately when my class is cancelled or the room changes, so I don't go to the wrong place.
US-03	As a student, I want to check official announcements in one place so I don't have to monitor multiple Telegram groups.
US-04	As a student, I want to search for a lecturer's contact information directly in the app so I don't have to go through the class representative.
US-05	As a student, I want to view my exam schedule in the app so I never miss an exam due to confusion about dates.
US-06	As a student, I want the app to work offline so I can check my timetable even when I have no internet connection.
US-07	As a student, I want to find a building or room on the campus map so I don't get lost looking for an office.

9.2 Staff User Stories

ID	User Story
US-08	As a lecturer, I want to post a class cancellation in 3 steps or fewer so I can notify students even when I am busy.
US-09	As a staff member, I want to push an announcement to all students in a department so I don't have to rely on class representatives.
US-10	As an administrator, I want to update room information and office hours in the directory so students always have accurate contact details.

10. Summary of Findings

The research confirms that the current campus information system is fundamentally broken. Students and staff have adapted by creating informal channels Telegram groups, class representatives, word-of-mouth — that partially compensate for the absence of an official digital system. These workarounds fail regularly, with documented consequences including failed assessments and grade losses.

The Smart Campus Companion addresses every identified failure point:

Problem (From Research)	Solution (In App)
No centralized information source	Single official app replaces fragmented Telegram channels
Dependence on class representatives	Staff post directly to students via admin dashboard
No real-time alerts	Push notifications for cancellations, room changes, and announcements
Cannot find staff contact info	Searchable staff and office directory built into the app
Campus navigation difficulty	Interactive campus map with building and room search
No offline access	Offline-first architecture syncs content for use without internet

All 3 staff respondents confirmed willingness to use a simple admin dashboard. 90% of students have been affected by information failures. The demand is real, the pain is documented, and the solution is validated.

End of Document